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Can news improve Nasdaq 100 forecasts?

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Motivation

Nasdaq-100 tracks **100 large non-financial companies**
including well-known companies like Apple, Microsoft, Nvidia, Amazon, and Meta



Many factors influence the index
(Actual correlation are unknown and constantly changes)



News may contain **useful signals** about these movements



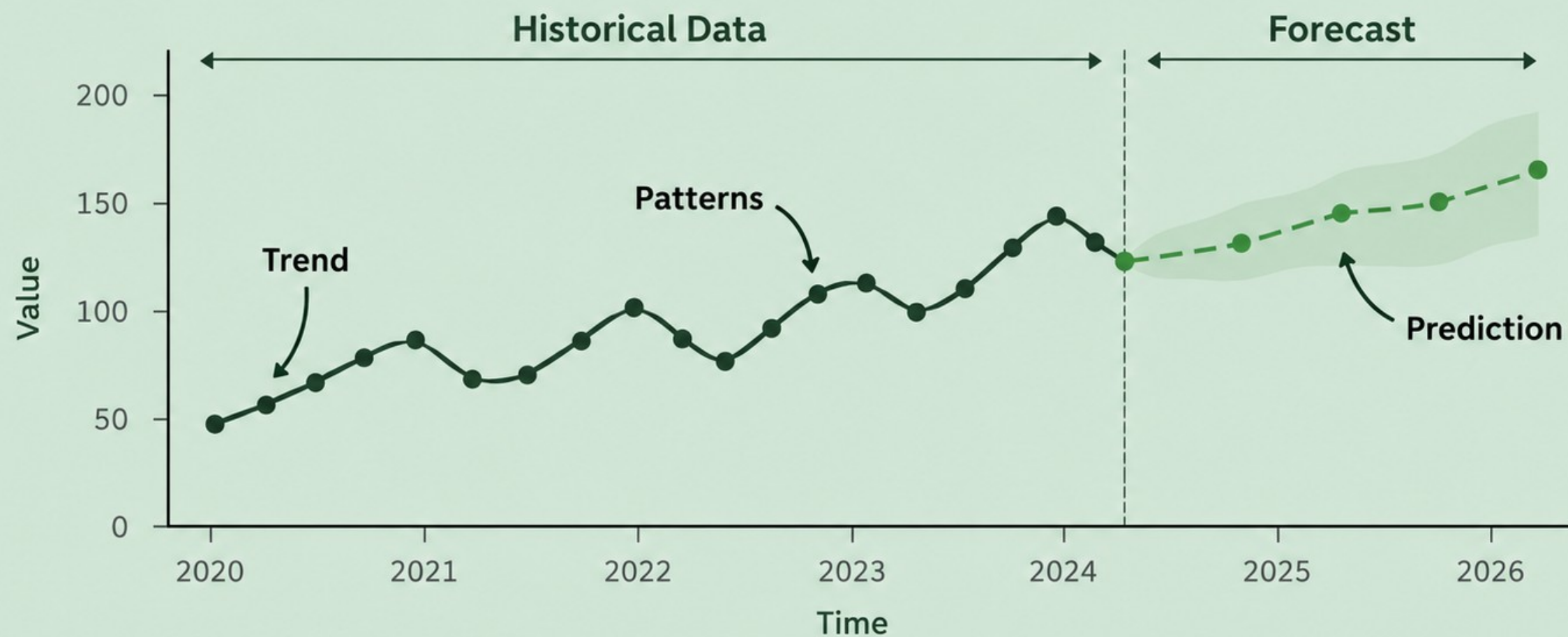
Can a **deep-learning model** understand and use these relationships?

Existing Work

- 2018–2022: **LSTM/GRU, CNN-LSTM** and **attention-based recurrent models** dominated financial forecasting.
- 2021–2024: **Transformers** and **specialized time-series** architectures arrived: TFT, Informer/Autoformer, PatchTST, TimesNet, iTransformer and TimeMixer.
- 2024–2026: The field moved toward **time-series foundation models** such as Chronos, TimesFM and Moirai, along with finance-specific pretraining, **graph neural networks**, text/news models such as FinBERT, regime-aware models and probabilistic forecasting.

Time Series Models

sequence of observations recorded over time | Order matters



Chronos-2

- A pretrained time-series foundation model
- Supports additional inputs
- Can be efficiently fine-tuned with LoRA
- Open source
- Pretrained with various data

119.5M base parameters

512-session context

21 forecast quantiles

General Market Data

- **QQQ history:** Returns, overnight gaps, volume, momentum, and volatility.
- **Related markets:** Nasdaq volatility (VXN), Treasury bonds (TLT), and corporate bonds (HYG).
- **Macroeconomics:** Interest rates, yield curves, real yields, and inflation.
- **Financial Events:** Fed meetings, inflation releases, employment reports, and options expiration.
- **External signals:** Economic-policy uncertainty, Fed statement tone, and SEC filing activity.

News Data - GDELT

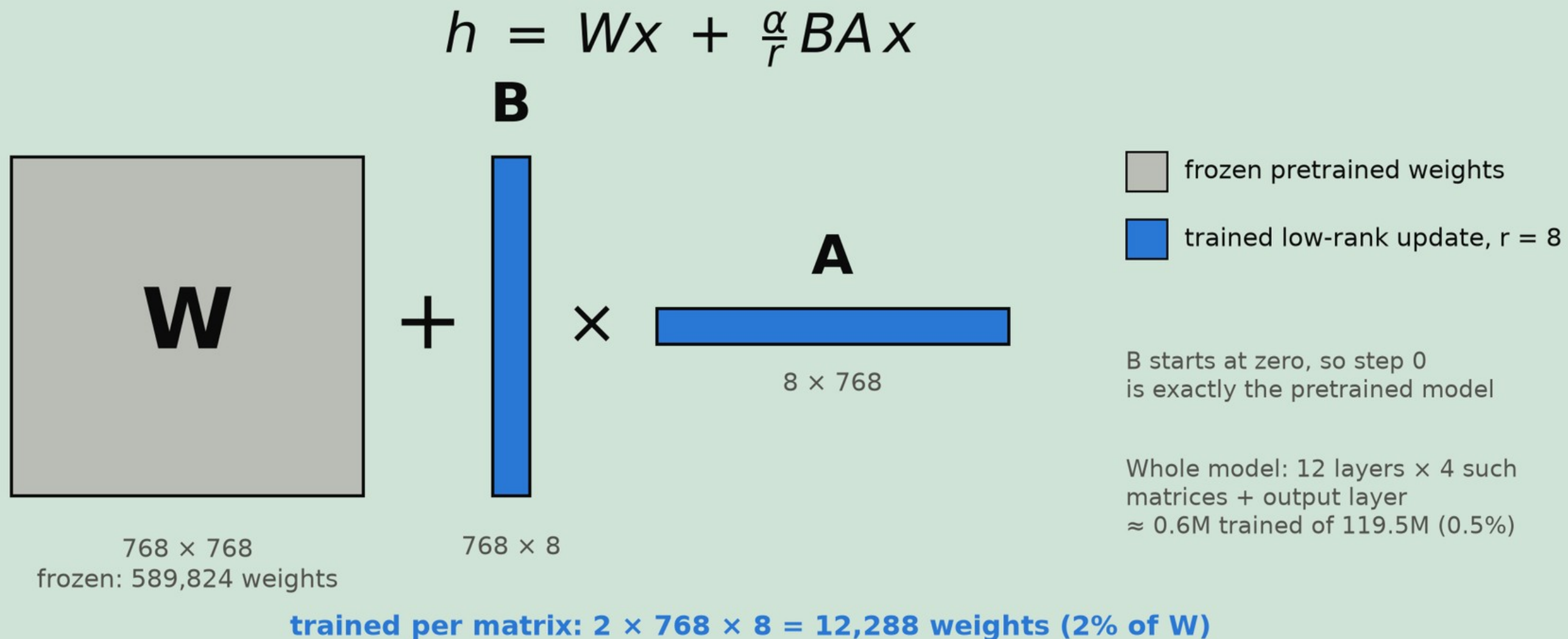
- A free open platform that tracks global news and converts it into data on topics, events, and tone.
- GDELT delivers news not just as text, but as structured daily numbers that a model can use directly.

We focus on six topics:

- AI
- Semiconductors
- Federal Reserve policy
- Inflation
- big-tech earnings
- recession

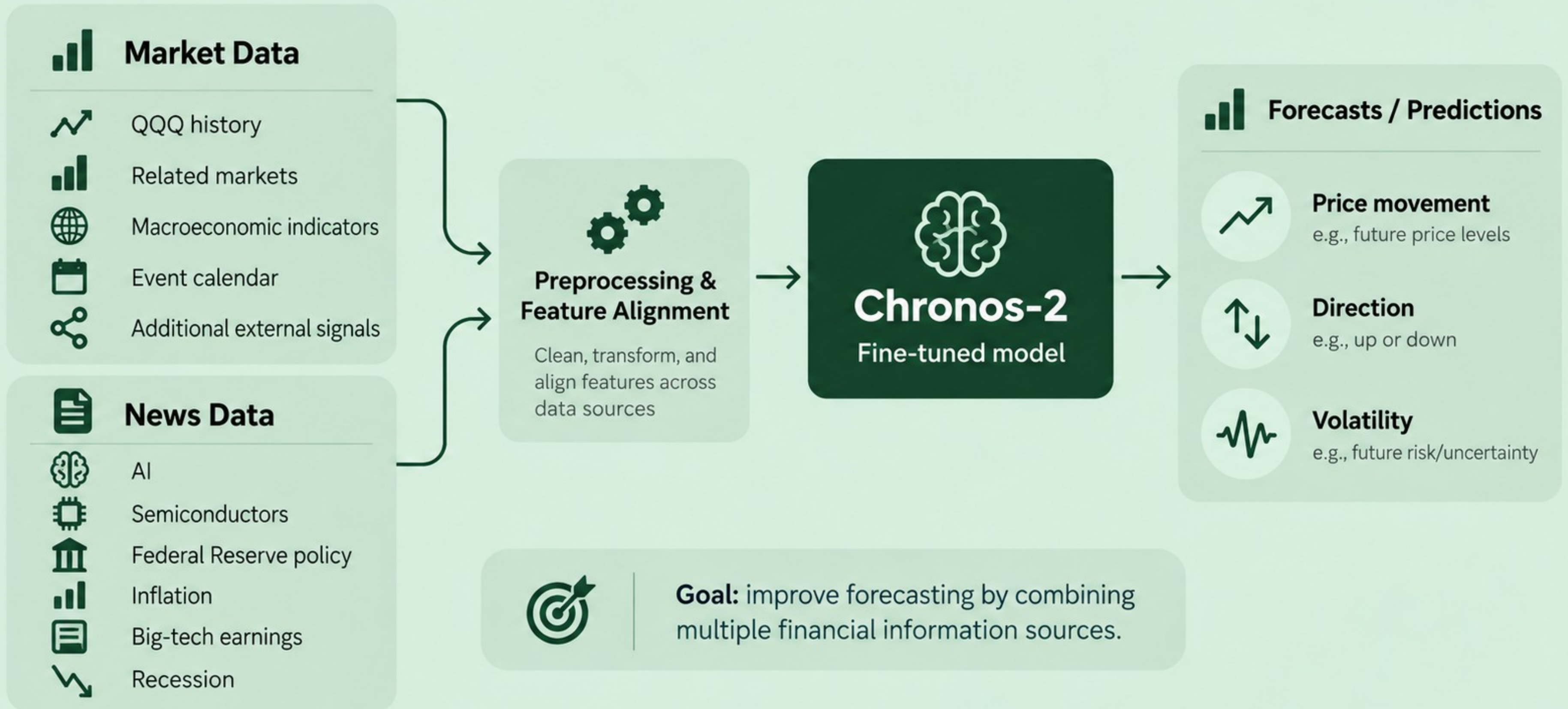
LoRA - Low-Rank Adaptation

fine-tune a large pretrained model without updating all of its weights



Fine-Tuning Chronos-2

Using market data and financial data together



Training and Evaluation Process

Two experiment rounds:

An initial 18-run source comparison

Followed by 108 fine-tuning runs covering four evaluation years and three random seeds.



After training:

We freeze the weights, forecast five separate daily returns at each origin, and advance five trading sessions.

How we score a forecast

The model outputs 21 quantiles for each of the next five days

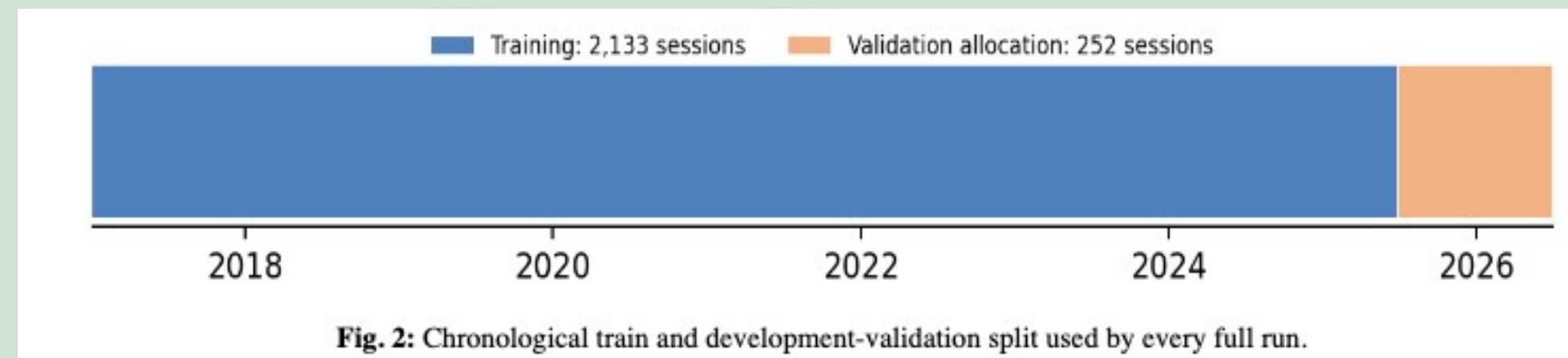
Weighted quantile loss: rewards the whole distribution, lower is better

Coverage: the 10 to 90 percent band should contain 80 percent of outcomes

**Direction is compared with “always up”, not with 50 percent:
in 2025, always-up scores 57 percent**

Fine Tune Run

- The model receives observations through the previous QQQ close and produces forecast for each of the next five trading-day returns.
- After those five realized sessions, the origin advances by five sessions. Fifty non-overlapping origins therefore score 250 daily outcomes; the final two sessions of the 252-session allocation cannot form a complete block
- Fine tune with different news source separately and compare to control
- Check both direction and index price predictions



$$\text{WQL reduction} = \frac{\text{Control WQL} - \text{News-model WQL}}{\text{Control WQL}} \times 100$$

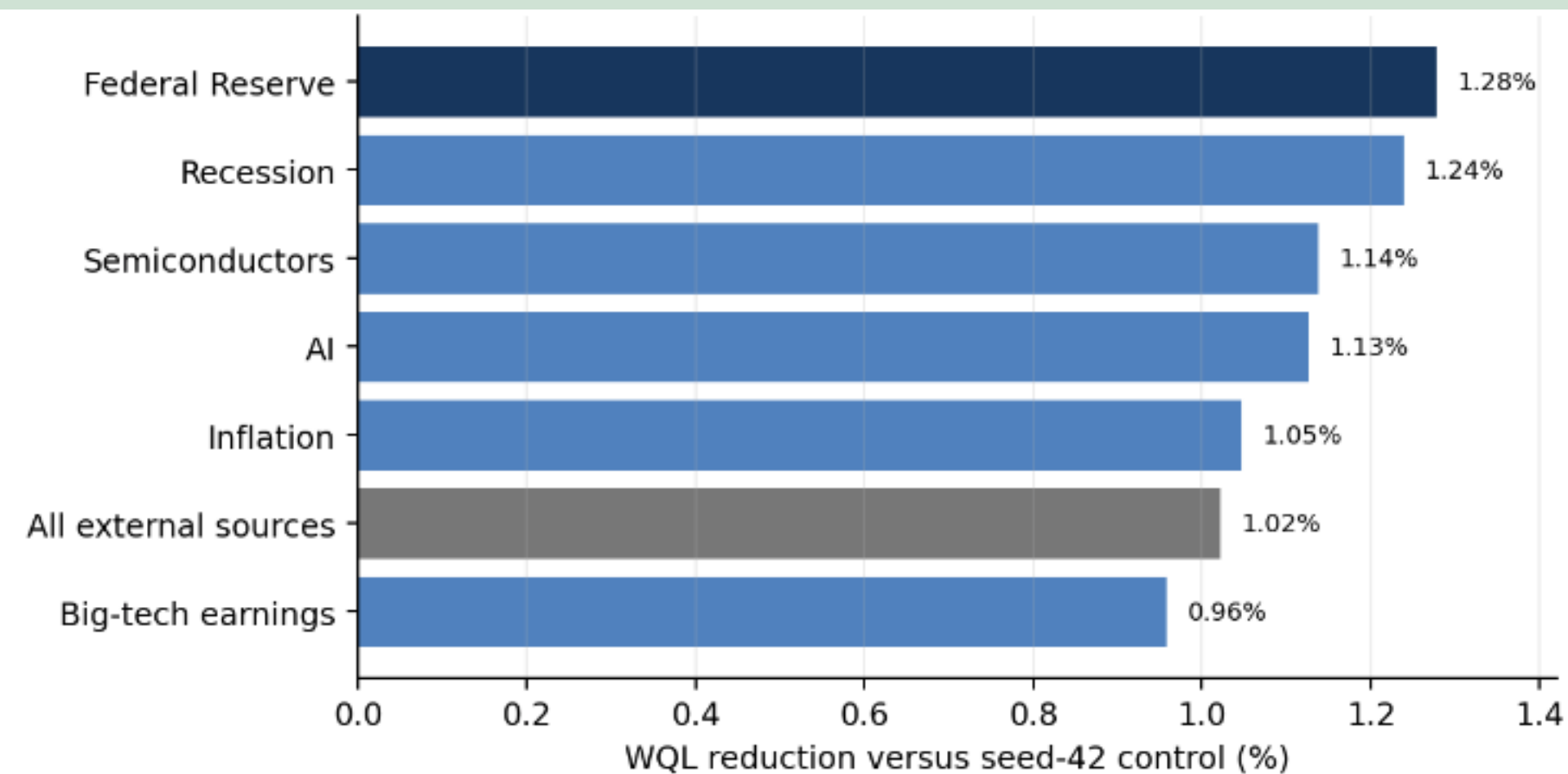


Fig. 5: WQL reduction versus the matched seed-42 control. The grey bar is the previous all-external model.

Rank	News data	WQL	Reduction	Correct direction
1	Federal Reserve	0.686526	1.28%	144 / 250
2	Recession	0.686796	1.24%	144 / 250
3	Semiconductors	0.687499	1.14%	144 / 250
4	AI	0.687580	1.13%	143 / 250
5	Inflation	0.688131	1.05%	142 / 250
6	All external sources	0.688303	1.02%	145 / 250
7	Big-tech earnings	0.688750	0.96%	142 / 250
-	Control: no news	0.695413	-	140 / 250

Table 4: Eight-case topic comparison on the same 250 daily outcomes.

Fine Tune Run

B How we find what the model uses

Take one input. Replace its 512 days with the same input from a different random week. Forecast again.

Forecast got worse → the model was using it

No change → ignored

Forecast got better → the input was misleading it

42 inputs - One surprising standout

News signals

Mostly close to zero → little evidence of meaningful use

VXN change

Large negative effect → Removing it improves the forecast

The pretrained model doesn't strongly exploit our news signals.
Instead, it reacts strongly to volatility.



Does fine-tuning change what the model uses?

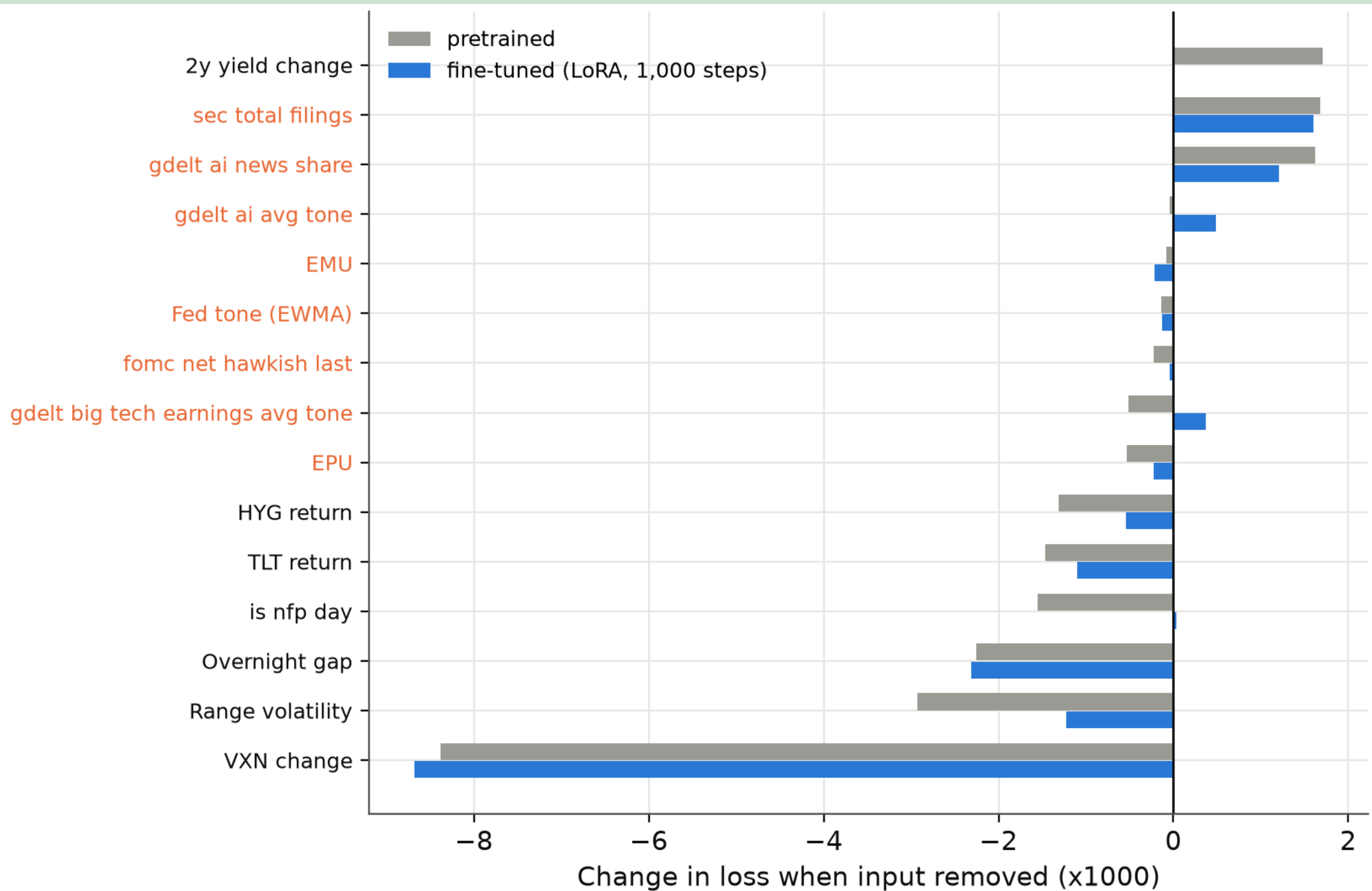
**The model's input-usage pattern stays remarkably
similar**

0.91 correlation

Then underneath:

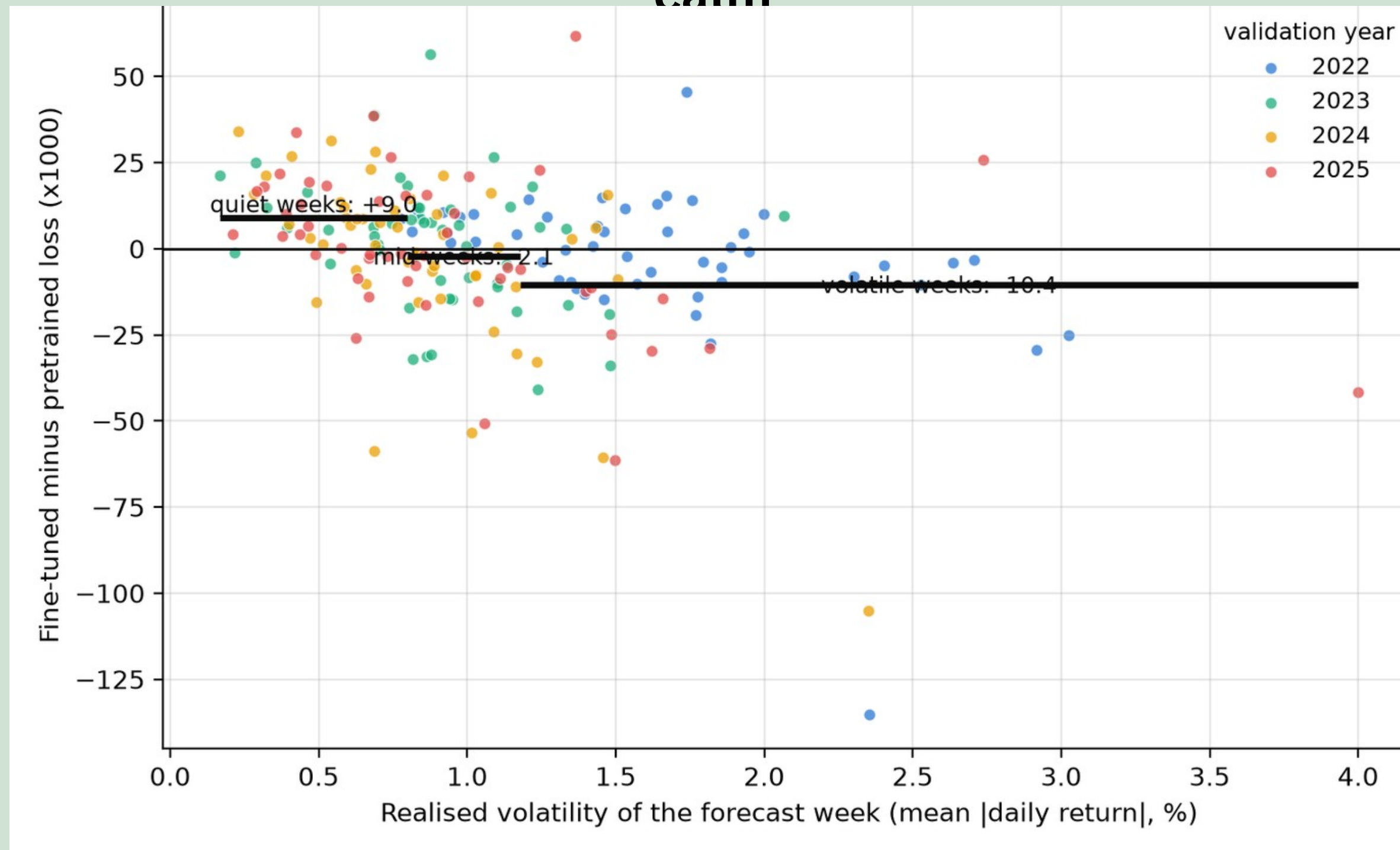
VXN: still misleading

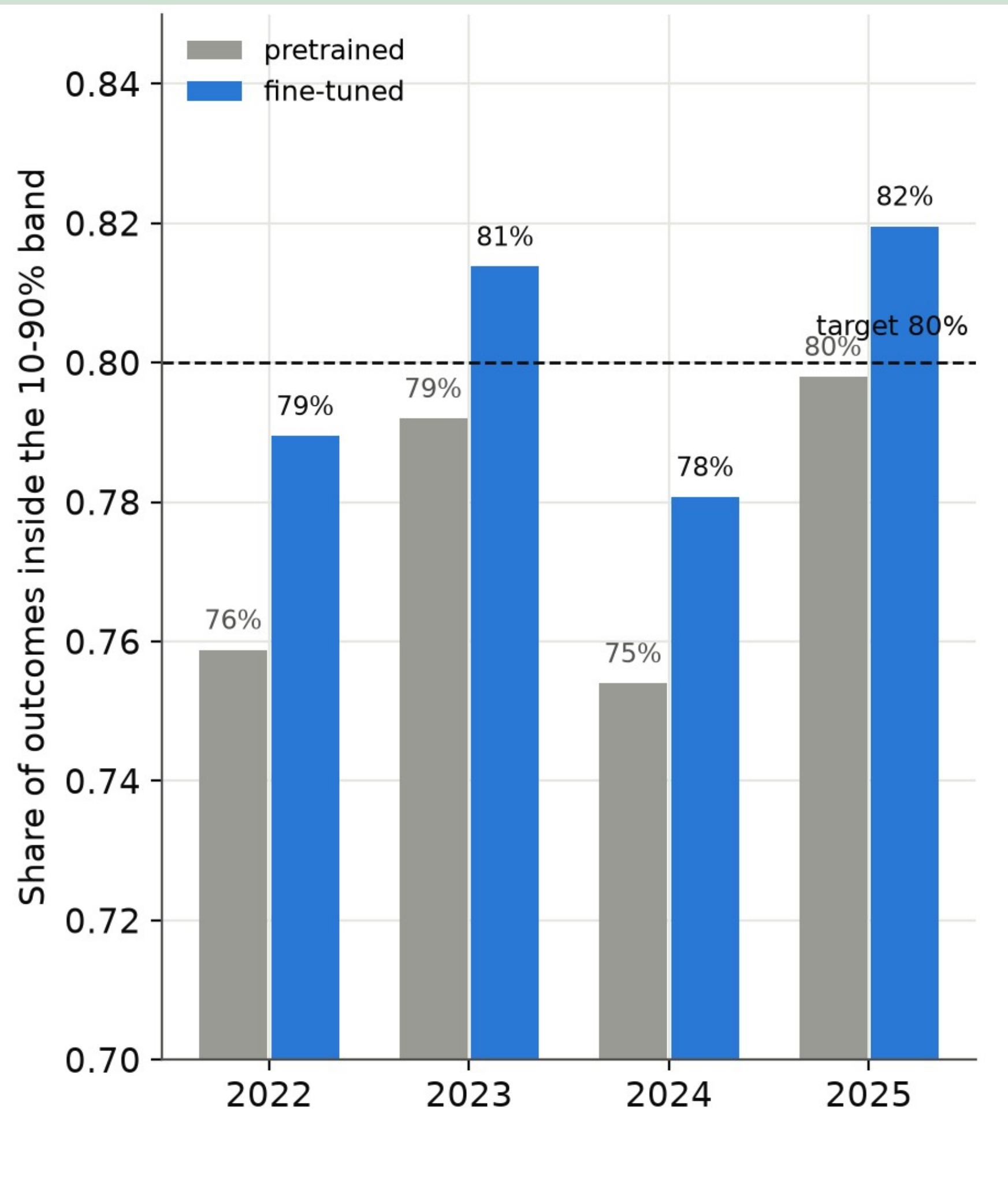
News: little evidence of meaningful use



What does fine-tuning change?

Fine-tuning helps when markets are volatile - but hurts when they're calm





Conclusions

- **No particular news source improved Nasdaq-100 forecasting**
- **Fine-tuning learns the scale of returns, not the meaning of the inputs: it widens the bands**
- **Fast market inputs like the VIX change hurt the pretrained model, and fine-tuning does not fix that**

Future work

- Full fine-tuning with a larger budget
- Longer news history and richer text
- Improve how we represent news.
- Compare news volume and average tone with richer features, such as unexpected changes in coverage, headline novelty, or relevance to major Nasdaq-100 companies.